Unit 10: Quadratic Graphs & Completing the Square

Student: SA
Tutor: Rachel Eglash

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Session 10.1:
Quadratic Graphs (Vertex, Axis, Intercepts)

Quick Reference: Quadratic Functions

Standard Form of a Quadratic

$y = ax^2 + bx + c$ where a, b, and c are constants and $a \neq 0$

Key Features of a Parabola

Vertex: The highest or lowest point on the parabola

Axis of Symmetry: The vertical line through the vertex

Formula: $x = -\frac{b}{2a}$

Direction:

If $a > 0$, parabola opens upward (vertex is minimum)

If $a < 0$, parabola opens downward (vertex is maximum)

y**-intercept:** Where the graph crosses the y-axis

Set $x = 0$, solve for y. Point: $(0, c)$

x**-intercepts:** Where the graph crosses the x-axis

Set $y = 0$, solve for x (factor or use other methods)

Finding the Vertex from Standard Form

Given $y = ax^2 + bx + c$:

Step 1: Find x-coordinate of vertex: $x = -\frac{b}{2a}$

Step 2: Substitute this x-value into the equation to find y

Step 3: Vertex is at point $\left(-\frac{b}{2a}, y\right)$

Homework 10.1:
Quadratic Graphs — Vertex, Axis, Intercepts

Part A: Identify Key Features

For each quadratic function, identify:

- Direction (opens up or down)
- y-intercept
- Axis of symmetry
- Vertex

Show your work for finding the vertex.

Problem 1:

$y = x^2 + 6x + 5$

Direction: ______________________

y-intercept: ______________________

Axis of symmetry: ______________________

Work for vertex:

Vertex: ______________________

Problem 2:

$y = -x^2 + 4x - 3$

Direction: __________________________

y-intercept: __________________________

Axis of symmetry: __________________________

Work for vertex:

Vertex: __________________________

Problem 3:

$y = 2x^2 - 8x + 3$

Direction: __________________________

y-intercept: __________________________

Axis of symmetry: __________________________

Work for vertex:

Vertex: __________________________

Part B: Find x-intercepts

Find the x-intercepts by setting $y = 0$ and factoring.

Problem 4:

$y = x^2 + 6x + 5$

x-intercepts: ____

Problem 5:

$y = x^2 - 9$

x-intercepts: ____

Problem 6:

$y = x^2 + 5x + 6$

x-intercepts: ______________________________

Problem 7:

$y = -x^2 + 2x + 8$

x-intercepts: ______________________________

Part C: Graph from Key Features

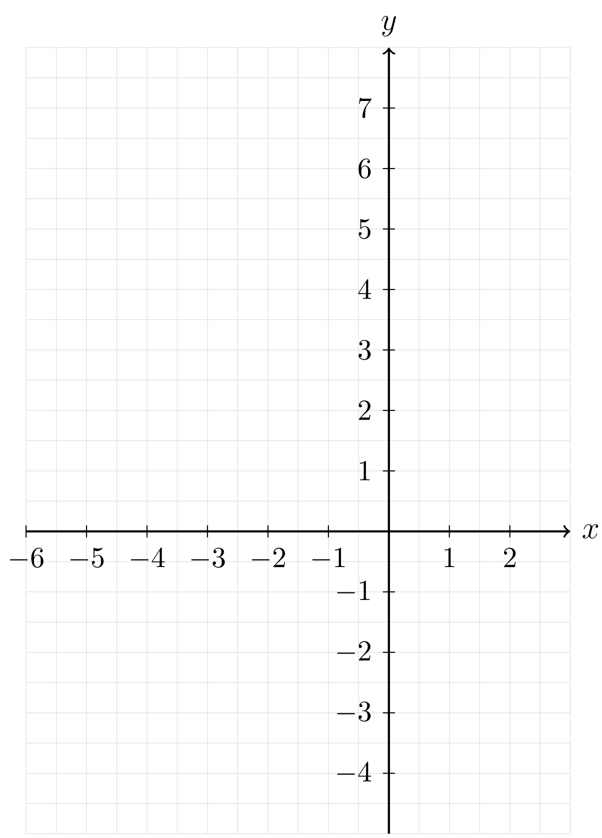
Use the key features you found to sketch a graph.

For each function:

- Plot the vertex
- Draw the axis of symmetry
- Plot the y-intercept
- Plot the x-intercepts (if you found them)
- Sketch the parabola

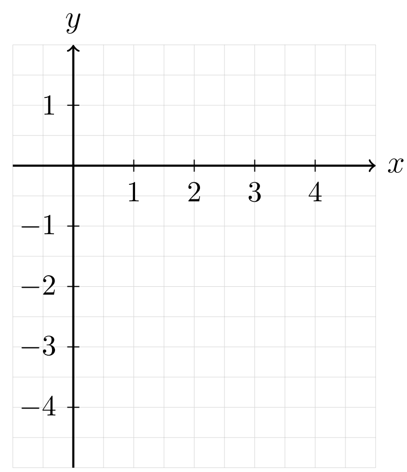
Problem 8:

Graph $y = x^2 + 6x + 5$ using your work from Problems 1 and 4.



Problem 9:

Graph $y = -x^2 + 4x - 3$ using your work from Problem 2.



Part D: Complete Analysis

For the following functions, find all key features AND sketch the graph.

Problem 10:

$y = x^2 - 4x$

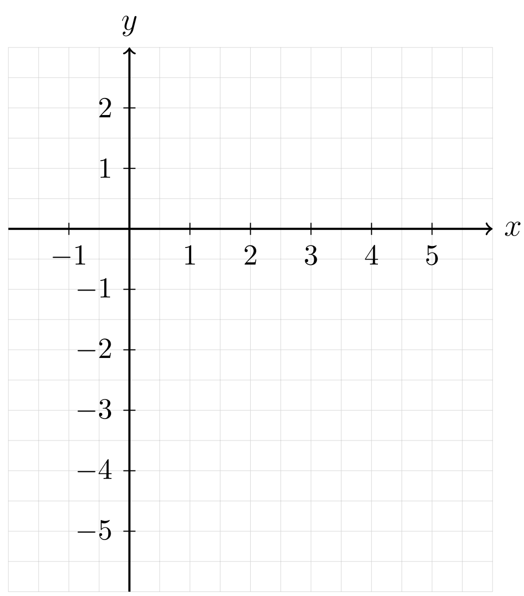
Direction: ______________________________

y-intercept: ______________________________

Axis of symmetry: ______________________________

Vertex: ______________________________

x-intercepts: ______________________________



Part E: Word Problems

Problem 11:

A toy rocket is launched from the ground.

Its height h in feet after t seconds is given by:

$h = -16t^2 + 64t$

Part A:

Find the axis of symmetry.

Part B:

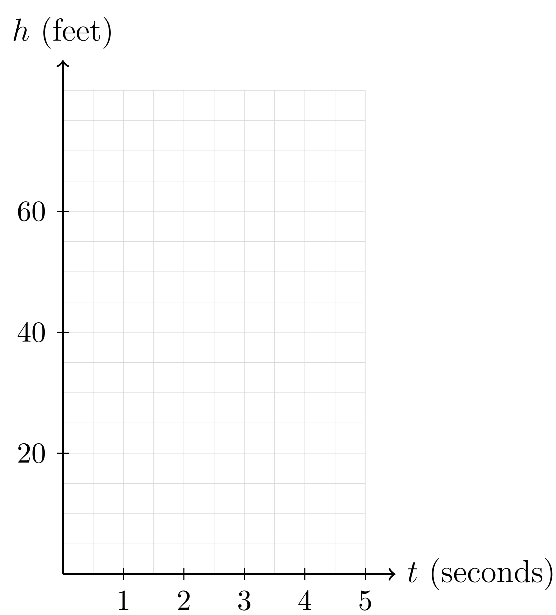
Find the vertex. What does the vertex represent in this context?

Part C:

Find the t-intercepts. What do they represent in this context?

Part D:

Sketch the graph of the rocket's height.



Problem 12:

A ball is thrown from a height of 6 feet.

Its height h in feet after t seconds is given by:

$h = -16t^2 + 32t + 6$

Part A:

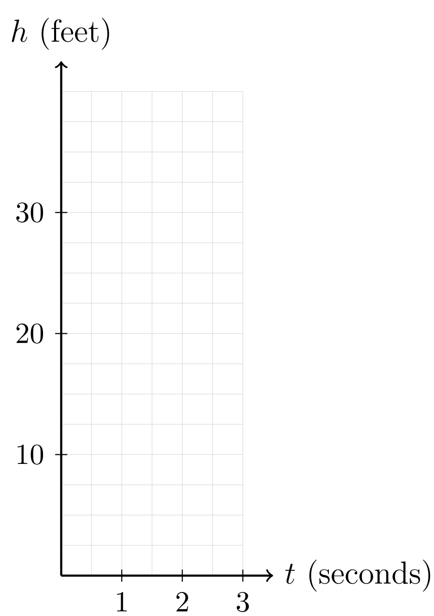
Find the maximum height of the ball and when it occurs.

Part B:

What is the initial height of the ball (when $t = 0$)?

Part C:

Sketch the graph showing the ball's path.



Session 10.2:
Solve by Square Roots; Completing the Square Setup

Quick Reference: Solving by Square Roots

Square Root Property

If $x^2 = k$, then $x = \pm\sqrt{k}$

Remember: When you take the square root, you get both positive and negative solutions!

Solving $(x - h)^2 = k$

Steps:

1. Take the square root of both sides: $x - h = \pm\sqrt{k}$
2. Solve for x: $x = h \pm \sqrt{k}$
3. Write two solutions: $x = h + \sqrt{k}$ or $x = h - \sqrt{k}$

Introduction to Completing the Square

Goal: Rewrite $x^2 + bx$ as a perfect square trinomial

Perfect Square Trinomial Pattern:

$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$

Key Idea: Take half of the x-coefficient, then square it.

To complete the square for $x^2 + bx$:

1. Find $\frac{b}{2}$
2. Square it: $\left(\frac{b}{2}\right)^2$
3. Add this value to create: $x^2 + bx + \left(\frac{b}{2}\right)^2$
4. Factor as: $\left(x + \frac{b}{2}\right)^2$

Homework 10.2:
Solve by Square Roots; Completing the Square Setup

Part A: Solve Using Square Roots

Solve each equation. Give exact answers (leave in square root form).

Problem 1:

$x^2 = 25$

Problem 2:

$x^2 = 18$

Problem 3:

$x^2 - 49 = 0$

Problem 4:

$3x^2 = 75$

Part B: Solve $(x-h)^2 = k$

Solve each equation. Give exact answers.

Problem 5:

$(x-3)^2 = 16$

Problem 6:

$(x+5)^2 = 9$

Problem 7:

$(x-2)^2 = 7$

Problem 8:

$(x+1)^2 - 20 = 0$

Part C: Complete the Square (Setup Only)

For each expression, find the value needed to complete the square.

Then write the perfect square trinomial and its factored form.

Problem 9:

$x^2 + 8x+$ ____________

Value to add: ________________________

Perfect square trinomial: ________________________

Factored form: ________________________

Problem 10:

$x^2 - 10x+$ ____________

Value to add: ________________________

Perfect square trinomial: ________________________

Factored form: ________________________

Problem 11:

$x^2 + 6x +$ ____________

Value to add: ________________________

Perfect square trinomial: ________________________

Factored form: ________________________

Problem 12:

$x^2 - 14x +$ ____________

Value to add: ________________________

Perfect square trinomial: ________________________

Factored form: ________________________

Part D: Word Problem

Problem 13:

A square picture frame has an area of 80 square inches.

Part A:

Write an equation for the side length s in inches.

Part B:

Solve for s using square roots. Give both the exact and approximate answer (round to one decimal place).

Part C:

Which solution makes sense in this context?

Session 10.3:
Completing the Square to Solve; Vertex Form

Quick Reference: Completing the Square to Solve

Steps to Solve $ax^2 + bx + c = 0$ by Completing the Square

Step 1: Move constant to the right side

$ax^2 + bx = -c$

Step 2: If $a \neq 1$, divide everything by a

$x^2 + \frac{b}{a}x = -\frac{c}{a}$

Step 3: Complete the square on the left side

- Take half of the x-coefficient: $\frac{1}{2} \cdot \frac{b}{a} = \frac{b}{2a}$
- Square it: $\left(\frac{b}{2a}\right)^2$
- Add to both sides

Step 4: Factor the left side as a perfect square

Step 5: Solve using square roots

Converting to Vertex Form

Standard form: $y = ax^2 + bx + c$

Vertex form: $y = a(x - h)^2 + k$ where (h, k) is the vertex

Process:

1. If $a \neq 1$, factor out a from the x-terms only
2. Complete the square inside the parentheses
3. Simplify to get vertex form

Reading the vertex: In $y = a(x - h)^2 + k$, the vertex is (h, k)

Note the sign! $y = (x - 3)^2 + 5$ has vertex $(3, 5)$, not $(-3, 5)$

Homework 10.3:
Completing the Square to Solve; Vertex Form

Part A: Solve by Completing the Square

Solve each equation by completing the square. Show all steps.

Problem 1:

$x^2 + 6x + 5 = 0$

Problem 2:

$x^2 - 8x + 12 = 0$

Problem 3:

$x^2 + 4x - 5 = 0$

Problem 4:

$x^2 + 10x = -9$

Part B: Solve When $a \neq 1$

Solve by completing the square. Remember to divide by a first!

Problem 5:

$2x^2 + 8x - 10 = 0$

Problem 6:

$3x^2 - 12x + 9 = 0$

Part C: Convert to Vertex Form

Rewrite each function in vertex form by completing the square.

Then identify the vertex.

Problem 7:

$y = x^2 + 8x + 10$

Vertex form: ____________________________________

Vertex: ________________________

Problem 8:

$y = x^2 - 6x + 5$

Vertex form: __________________________________

Vertex: _______________________

Problem 9:

$y = x^2 + 4x - 1$

Vertex form: __________________________________

Vertex: _______________________

Part D: Convert When $a \neq 1$

Rewrite in vertex form. Factor out a from the x-terms first!

Problem 10:

$y = 2x^2 + 8x + 3$

Vertex form: ____________________________________

Vertex: ________________________

Problem 11:

$y = -x^2 + 6x - 5$

Vertex form: ________________________________

Vertex: ______________________

Part E: Word Problem

Problem 12:

A ball is thrown upward from a height of 4 feet.

Its height h in feet after t seconds is given by:

$h = -16t^2 + 32t + 4$

Part A:

Rewrite the equation in vertex form by completing the square.

Part B:

What is the maximum height of the ball? At what time does it reach this height?

Part C:

Explain what the vertex represents in this context.

Unit 10 Quiz: Quadratic Graphs & Completing the Square

Section 1: Key Features of Quadratics

For the function $y = x^2 - 4x + 3$, find:

Problem 1:

Direction (opens up or down)

Problem 2:

y-intercept

Problem 3:

Axis of symmetry

Problem 4:

Vertex (show your work)

Section 2: Solve by Square Roots

Solve each equation. Give exact answers.

Problem 5:

$x^2 = 36$

Problem 6:

$(x - 4)^2 = 25$

Problem 7:

$(x + 2)^2 = 12$

Section 3: Complete the Square

Problem 8:

What value completes the square for $x^2 + 10x+$ ____________ ?

Write the perfect square trinomial and factor it.

Section 4: Solve by Completing the Square

Problem 9:

Solve $x^2 + 8x + 7 = 0$ by completing the square. Show all steps.

Section 5: Vertex Form

Problem 10:

Rewrite $y = x^2 + 6x + 4$ in vertex form.

Problem 11:

What is the vertex of $y = (x - 5)^2 + 3$?

Section 6: Word Problem

Problem 12:

A diver jumps from a platform.

Her height h in feet above the water after t seconds is:

$h = -16t^2 + 8t + 24$

Find the maximum height of the diver and the time when she reaches it.